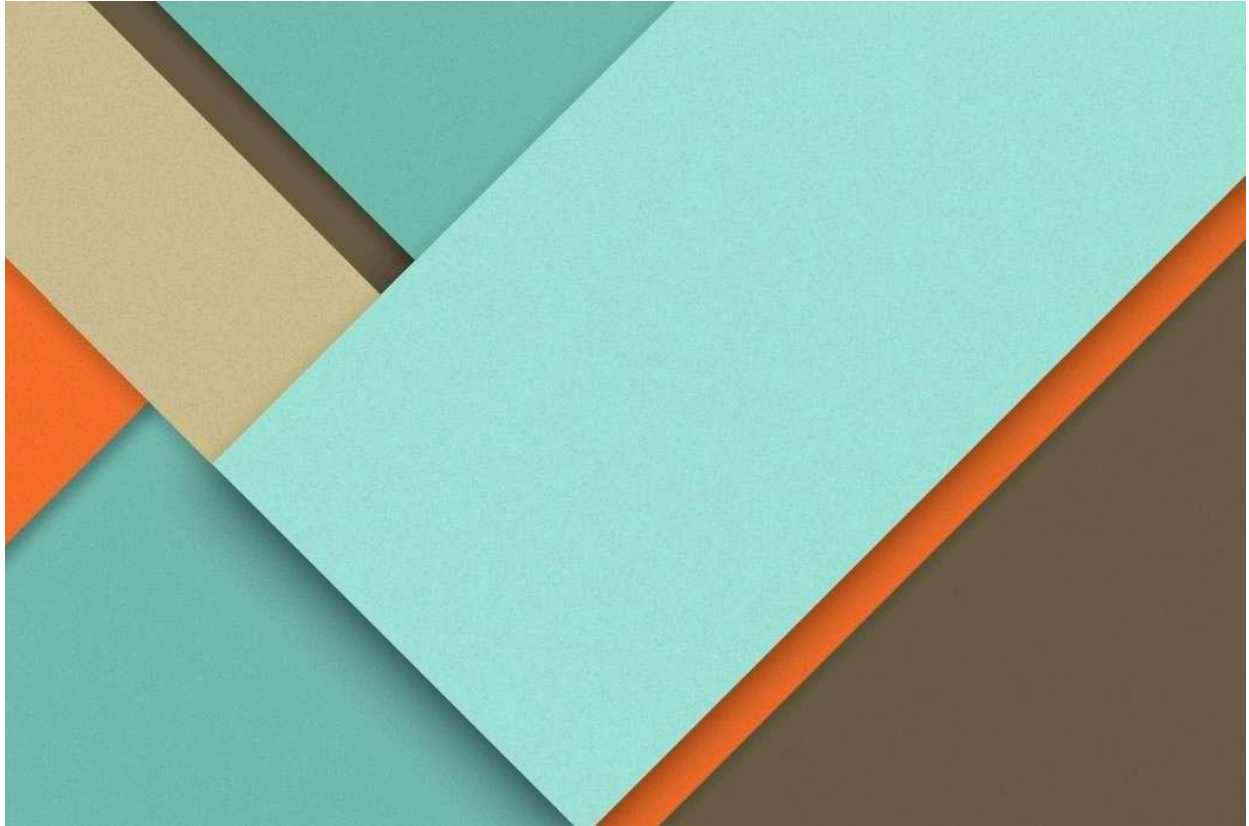




CLOUD COMPUTING

LAB-8

24-03-2025

GITHUB: <https://github.com/akshayag2005/CLOUD-COMPUTING>

AKSHAY

PES2UG23CS044

CSE-A

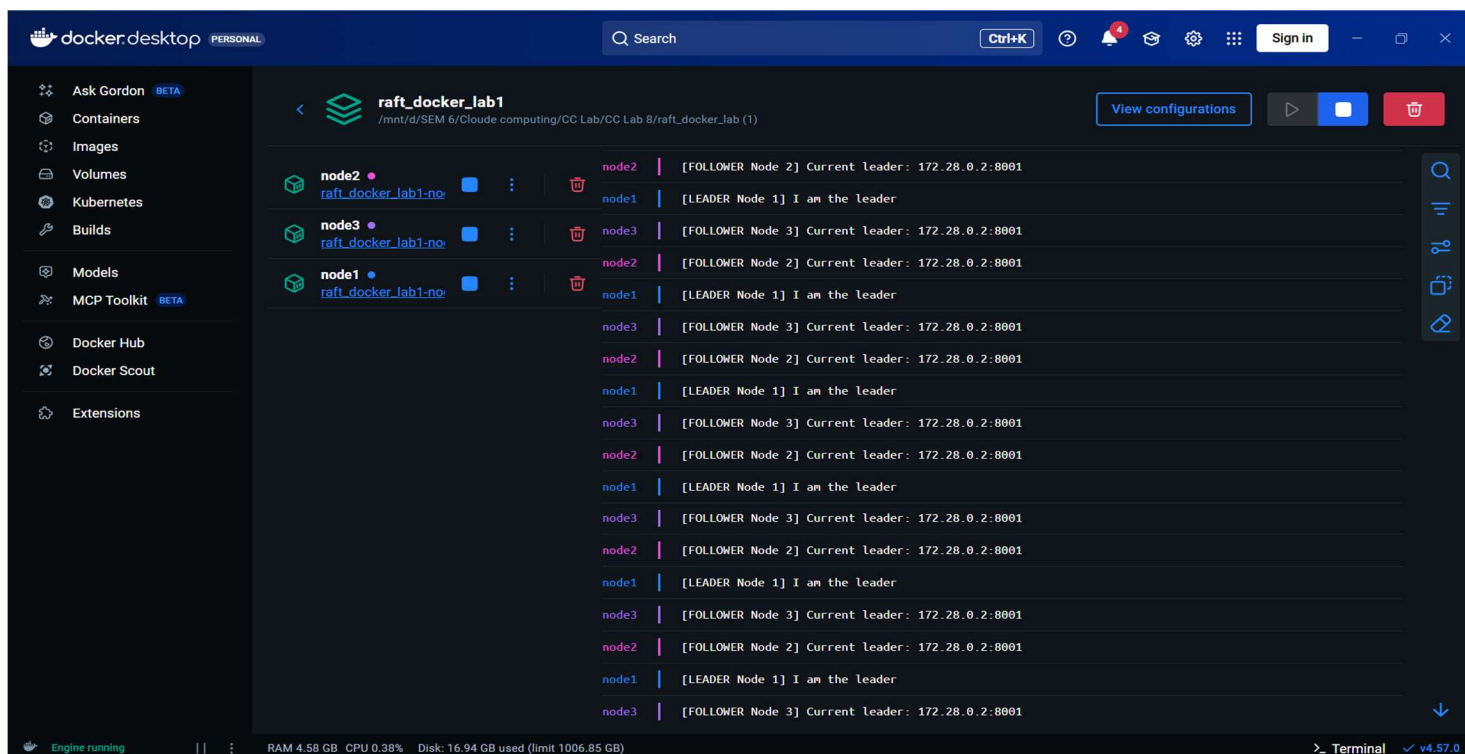
Screenshot 1:

```
(akshay@AKSHAY) - [mnt/d/SEM 6/Cloud computing/CC Lab/CC Lab 8/raft_docker_lab (1)]  
$ ls  
docker-compose.yml  Dockerfile  node.py  requirements.txt
```

Screenshot 2A:

```
=> => exporting manifest sha256:15d9d52ec916c3cf13fa53f3e2297dec92af424da8e4ca2002113b70d9add088 0.1s  
=> => exporting config sha256:3f4f329e3eb8bf25fb83f37d7c7fa0c3ce0970cd160a6761a13dc5cec7c8ccec 0.1s  
=> => exporting attestation manifest sha256:54abc08c1458e4631573513d008da548f5ca2dcf62c12ea56754ble02569f5ce 0.1s  
=> => exporting manifest list sha256:c69d8c13310012e3c05a4369656ebf2c661bb7bdb29c16f36fb243287c0baaec 0.1s  
=> => naming to docker.io/library/raft_docker_lab1-node3:latest 0.0s  
=> => unpacking to docker.io/library/raft_docker_lab1-node3:latest 1.1s  
=> [node2] resolving provenance for metadata file 0.1s  
=> [node3] resolving provenance for metadata file 0.1s  
=> [node1] resolving provenance for metadata file 0.0s  
[+] up 7/7  
✔ Image raft_docker_lab1-node3 Built 56.8s  
✔ Image raft_docker_lab1-node1 Built 56.8s  
✔ Image raft_docker_lab1-node2 Built 56.8s  
✔ Network raft_docker_lab1-raftnet Created 0.1s  
✔ Container node2 Created 0.3s  
✔ Container node1 Created 0.3s  
✔ Container node3 Created 0.3s  
Attaching to node1, node2, node3  
node2 | Starting Node 2 at 172.28.0.3:8002  
node2 | [Node 2] Waiting for leader election...  
node1 | Starting Node 1 at 172.28.0.2:8001  
node1 | [Node 1] Waiting for leader election...  
node3 | Starting Node 3 at 172.28.0.4:8003  
node3 | [Node 3] Waiting for leader election...
```

Screenshot 2B:




```
node1 | [LEADER Node 1] I am the leader
node3 | [FOLLOWER Node 3] Current leader: 172.28.0.2:8001
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.2:8001
```

[illegible][illegible]

[illegible]

CONCLUSIONS

[illegible]

Screenshot 4:

```
(akshay@AKSHAY) [/mnt/d/SEM 6/Cloude computing/CC Lab/CC Lab 8]
$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
824e5624f4c8	raft_docker_lab1-node2	"python node.py 2"	17 minutes ago	Up 17 minutes		node2
ce083b9f7aca	raft_docker_lab1-node3	"python node.py 3"	17 minutes ago	Up 17 minutes		node3
0bfb06de1e71	raft_docker_lab1-node1	"python node.py 1"	17 minutes ago	Up 17 minutes		node1
ddf8a4ee2fa8	safeher_app-motion_service	"python motion_servi..."	5 weeks ago	Restarting (2) 38 seconds ago		safeher_motion
68d86e5580e9	safeher_app-communication_service	"python mqtt_websock..."	5 weeks ago	Restarting (2) 38 seconds ago		safeher_communication

Screenshot 5:

```
(akshay@AKSHAY) [/mnt/d/SEM 6/Cloude computing/CC Lab/CC Lab 8/raft_docker_lab (1)]
$ docker network inspect raft_docker_lab1_raftnet
```

```
[
  {
    "Name": "raft_docker_lab1_raftnet",
    "Id": "0c68c25de22dc7b89e493add32f2ff5d3bfa721dd45001a37c3ec941c6c3d8e8",
    "Created": "2026-03-24T09:42:53.037502387Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.28.0.0/16",
          "IPRange": "",
          "Gateway": "172.28.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",

```

Screenshot 6A:

```
node2 | [Node 2] Leader changed: None
      | [Node 2] Waiting for leader election...
node3 | [Node 3] Leader changed: None
      | [Node 3] Waiting for leader election...
node2 | [Node 2] Leader changed: 172.28.0.4:8003
      | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node3 | [Node 3] Leader changed: 172.28.0.4:8003
      | [LEADER Node 3] I am the leader
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node3 | [LEADER Node 3] I am the leader
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node3 | [LEADER Node 3] I am the leader
```

Screenshot 6B:

```
[Node 2] Leader changed: None
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

Screenshot 6C:

```
[Node 3] Leader changed: None
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.4:8003
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
[LEADER Node 3] I am the leader
```

Screenshot 7A:

 node2  	 [Node 1] Leader changed: 172.28.0.4:8003 [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
 node3  	 [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
 node1  	 [LEADER Node 3] I am the leader
	 [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
	 [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
	 [LEADER Node 3] I am the leader
	 [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
	 [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
	 [LEADER Node 3] I am the leader
	 [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
	 [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
	 [LEADER Node 3] I am the leader
	 [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
	 [FOLLOWER Node 2] Current leader: 172.28.0.4:8003

Screenshot 7B:

```
[LEADER Node 1] I am the leader
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
[FOLLOWER Node 1] Current leader: 172.28.0.4:8003
```

Screenshot 8:

```
(akshay@AKSHAY) - [mnt/d/SEM 6/Cloude computing/CC Lab/CC Lab 8/raft_docker_lab (1)]
$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
7d4e9bbcb755	raft_docker_lab1-node4	"python node.py 4"	About a minute ago	Up About a minute		node4
5115a16075ca	raft_docker_lab1-node2	"python node.py 2"	About a minute ago	Up About a minute		node2
95728921a8d6	raft_docker_lab1-node3	"python node.py 3"	About a minute ago	Up About a minute		node3
0cf18357f665	raft_docker_lab1-node1	"python node.py 1"	About a minute ago	Up About a minute		node1
1aaa96a17626	raft_docker_lab1-node5	"python node.py 5"	About a minute ago	Up About a minute		node5
ddf8a4ee2fa8	safeher_app-motion_service	"python motion_servi..."	5 weeks ago	Restarting (2) 32 seconds ago		safeher_motion
68d86e5580e9	safeher_app-communication_service	"python mqtt_websock..."	5 weeks ago	Restarting (2) 32 seconds ago		safeher_communica tion

[illegible]

10

[illegible]

[illegible]

Screenshot 12:

[illegible]

Screenshot 13:

```
(akshay@AKSHAY)-[/mnt/d/SEM 6/Cloude computing/CC Lab/CC Lab 8/raft_docker_lab (1)]
$ docker logs -f node5
Starting Node 5 at 172.28.0.6:8005
[Node 5] Waiting for leader election...
[Node 5] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
```

Screenshot 14:

```
(akshay@AKSHAY)-[/mnt/d/SEM 6/Cloude computing/CC Lab/CC Lab 8/raft_docker_lab (1)]
$ docker stop node4
node4

(akshay@AKSHAY)-[/mnt/d/SEM 6/Cloude computing/CC Lab/CC Lab 8/raft_docker_lab (1)]
$ docker stop node5
node5

(akshay@AKSHAY)-[/mnt/d/SEM 6/Cloude computing/CC Lab/CC Lab 8/raft_docker_lab (1)]
$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
5115a16075ca	raft_docker_lab1-node2	"python node.py 2"	9 minutes ago	Up 9 minutes		node2
95728921a8d6	raft_docker_lab1-node3	"python node.py 3"	9 minutes ago	Up 9 minutes		node3
0cf18357f665	raft_docker_lab1-node1	"python node.py 1"	9 minutes ago	Up 9 minutes		node1
ddf8a4ee2fa8	safeher_app-motion_service	"python motion_servi..."	5 weeks ago	Restarting (2) 5 seconds ago		safeher_motion
68d86e5580e9	safeher_app-communication_service	"python mqtt_websock..."	5 weeks ago	Restarting (2) 6 seconds ago		safeher_communication

[illegible]